Composite orders: checking an alleged exception

The prime-field modular embedding-order conclusion extends to every integer $r > 2$ dividing the actual curve group order, provided the source supersingular trace condition and Hasse bound are retained. Primality of r is unnecessary in this arithmetic step. Hasse itself is an explicit premise; no general geometric classification is proved here. [pcc:unverified]

For prime $p \geq 5$, the source condition gives $p \mid t$. The premise $t^2 \leq 4p$ forces $t = 0$, so $\#E(\mathbf{F}_p) = p+1$. Consequently $p \equiv -1 \pmod{r}$. Its square is 1; if it were 1 already, then $r \mid 2$, contradicting $r > 2$. Thus its multiplicative order is 2, including composite r. The previous $r = 2$ counterexample remains a separate boundary. [pcc:unverified]

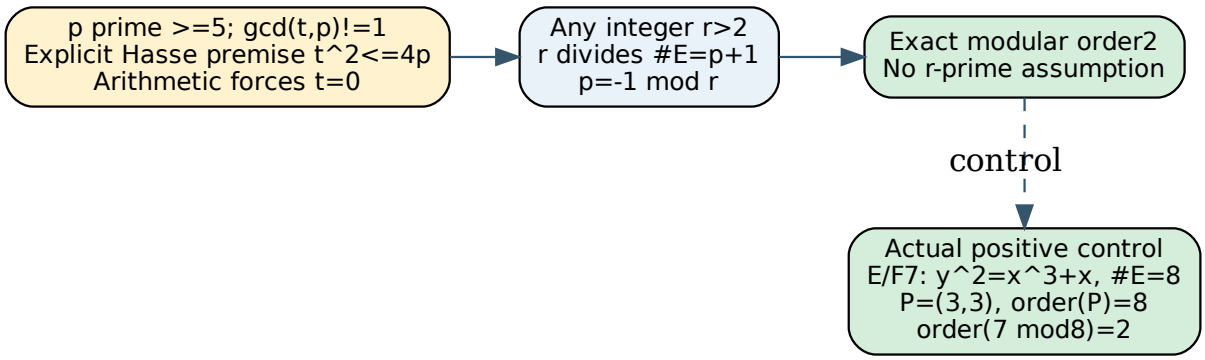

Figure 1: Checked arithmetic dependencies and actual composite positive control.

The actual smooth source curve $E/\mathbf{F}_7 : y^2 = x^3+x$ has 8 points and satisfies the source trace definition. The new actual point $P = (3, 3)$ obeys $8P = 0$ and $4P \neq 0$; Lean concludes addOrderOf$(P) = 8$. The modular embedding order of 7 modulo 8 is 2. These are actual Mathlib points, not an asserted cyclic operation table. [pcc:unverified]

Primary source: original FST2010, p230 Definition 2.1/Remark 2.2; integer subgroup order is allowed there. Lean source: ../../lean/CompositeTorsion.lean. No general cyclotomic-field or full geometric-torsion equivalence, binary/ternary census, or all-fourteen completion is inferred. No native population, timing or paid compute was run. [pcc:unverified]

Whole census axiom gate:205 declarations,0 failing,0 trusted-native exceptions, including generated declarations. Packet PCC-20261109-10789d binds the universal conditional theorem and unconditional actual control; producer standing remains unsigned. [pcc:unverified]